

Davide Donati

Curriculum Vitae

Via Vancini 7

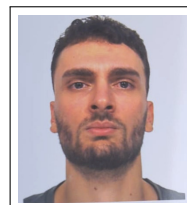
Bologna, Italy

+39 331 907 6926

donatidavide5@gmail.com

TucanoCode04

Driving License B, Own Car



Professional Experience

- September, 2025–Present **Member, Qubito**, Politecnico di Torino, Italy
- Student team dedicated to quantum computing: participation in hackathons, research on quantum technologies in general, and science outreach.
- 2022–2024 **Professional Volleyball Player**, Serie A, Italy
- Participation in international volleyball championships.(Lega Pallavolo Profile)
 - Development of skills in teamwork, discipline and stress management.
- 2019 **Work Placement**, Charity Shop, England
- Work experience in a charity shop.
 - Improved communication skills in an international context.

Education

- September, 2025–Present **Student**, Politecnico di Torino, Turin, Italy
- Master's degree in Quantum Engineering.
 - Comprehensive knowledge of the Master's programme in Quantum Engineering, with a particular interest in quantum computing and its applications.
- September, 2021–March, 2025 **Student**, University of Bologna, Bologna, Italy
- Bachelor's degree in Computer Science.
 - Comprehensive knowledge of the Computer Science programme, with a particular interest in Artificial Intelligence and Quantum Engineering.
- September, 2013–June, 2019 **High School Diploma**, Istituto Statale di Istruzione Superiore J.M. Keynes, Castel Maggiore (BO), Italy
- Specialization: Languages.
 - Languages studied: English, Spanish, German.

Computer Skills

- Programming
- Imperative Languages:**
 - C, C++:** Used for game development, thanks to user-controlled memory management, and to explore the architecture of these languages.
 - Java:** Experience in application development and in exploring optimization structures and algorithms.
 - Python:** Used both for scripting and for developing artificial intelligence applications, using libraries such as **TensorFlow** and **Keras**. Also used in quantum engineering with the **Qiskit** library.
 - Functional Languages:**
 - Haskell:** Used to explore structural recursion and understand the logic of a functional language.
 - Database Languages:**
 - SQL:** Experience using SQL to manage relational databases.
 - MongoDB:** Familiarity with non-relational databases for web application development.

- Web Development
 - **MERN Stack (MongoDB, Express, React, Node.js):** Experience in web application development:
 - **MongoDB:** NoSQL database for data management.
 - **Express:** Framework for building backends and managing the server.
 - **React:** Frontend library for building dynamic user interfaces.
 - **Node.js:** JavaScript runtime environment for the server.
 - **Web Standards:** Knowledge of HTML, CSS and JavaScript for creating interactive, cross-browser web pages. Knowledge of HTTP standards for handling requests and responses between client and server.
- Version Control and DevOps
 - **Git, GitHub, GitLab, GitKraken:** Used for version control, collaboration and code management across multiple projects.
 - **Docker:** Familiarity with containerization for developing scalable and portable applications.
- Software and Graphics
 - **Adobe Suite (Photoshop, Premiere Pro, After Effects):** Experience in graphic design, video editing and visual effects.
 - **Blender:** 3D modeling, rendering and animation.
 - **Unity:** Experience in game and simulation development.
 - **Krita:** Software for digital illustration and artwork.
 - **Microsoft Office (Word, Excel, PowerPoint):** Advanced knowledge for documentation, data analysis and professional presentations.

Languages

Italian	Native speaker	
English	Level C2	<i>Near-native proficiency</i>
Spanish	Level B2/C1	<i>Advanced proficiency</i>
German	Level B2	<i>Intermediate proficiency</i>

Personal Projects

- GitHub github.com/TucanoCode04 - All the projects listed below and many others, including study interests, are available on my GitHub.
- Main Projects
 - Thesis Project: Analysis of the differences between classical machine learning and quantum machine learning for missing data imputation in a dataset, using a quantum circuit to compare the performance and limits of quantum models against classical ones.
 - Qubit Electronics Lab Reports (Politecnico di Torino): work on **FPGAs**, analysis of basic electronic components and circuits, simulation of quantum dots (their charge and behavior), and use of **RFSoc**.
 - HPC Course Project (Politecnico di Torino): experiments on the Politecnico HPC cluster using **OpenMP**, **MPI** and **CUDA** to simulate waves and, above all, to stress-test and challenge the HPC system itself.
 - Selfie: Complete web application with a calendar, a study timer, a section to save and share notes, and a chat to coordinate study and plan group events.
 - Programming Game: An endless RPG in which players advance without limit through rooms and levels, acquiring upgrades to improve their abilities.
 - Chesscake: Web application to play several chess variants online, both against other players and against the computer, with user management and a global leaderboard.
 - Panda-OS: Operating system with multitasking, memory management and a basic interface, developed to explore fundamental architectures.
 - Image Deblur: Jupyter notebook project with **Keras** and **TensorFlow** to reconstruct sharp images from blurred versions using a U-Net network.
 - Projects in Unity, Blender, Premiere Pro and Photoshop: Development of games, 3D models, video editing and digital graphics.

Personal Interests

- Sports** Passionate about sports in general. I train regularly at the gym and have experience in powerlifting. I have also practiced combat sports such as judo and Brazilian jiu-jitsu, and played volleyball at a professional level.
- Travel** I have travelled extensively in Europe, initially thanks to my studies, which led me to live for a few weeks with host families in Germany, southern Spain and England (where I also worked). Later, I did a month of interrail, an experience that pushed me to keep travelling every year, both to discover new cultures and to visit museums and works of art, as well as to meet new people. I have visited Spain, Germany, the Netherlands, Denmark, the United Kingdom, Austria, Slovenia, Hungary, Poland, Croatia, Serbia, Macedonia, Greece and Romania.
- Professional Interests** Particular interest in quantum computing and machine learning, with a focus on applying advanced models to missing data imputation. More details and other projects on GitHub: github.com/TucanoCode04.